ID:	Name:

Set A

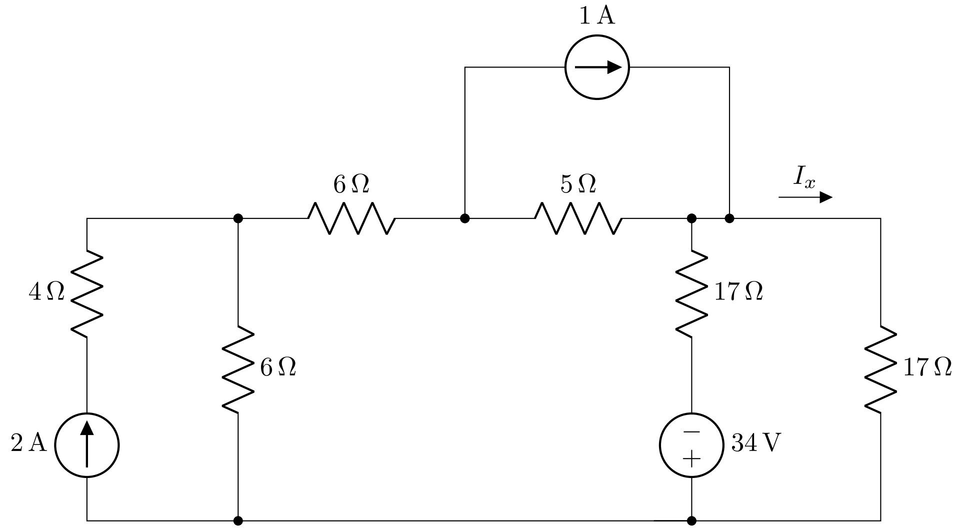
Brac University
Semester: Summer 2023
Course Code: CSE250
Circuits And Electronics

Assessment: *Makeup Midterm*
Duration: 1 hour 30 minutes
Date: August 12, 2023
Full Marks: 55

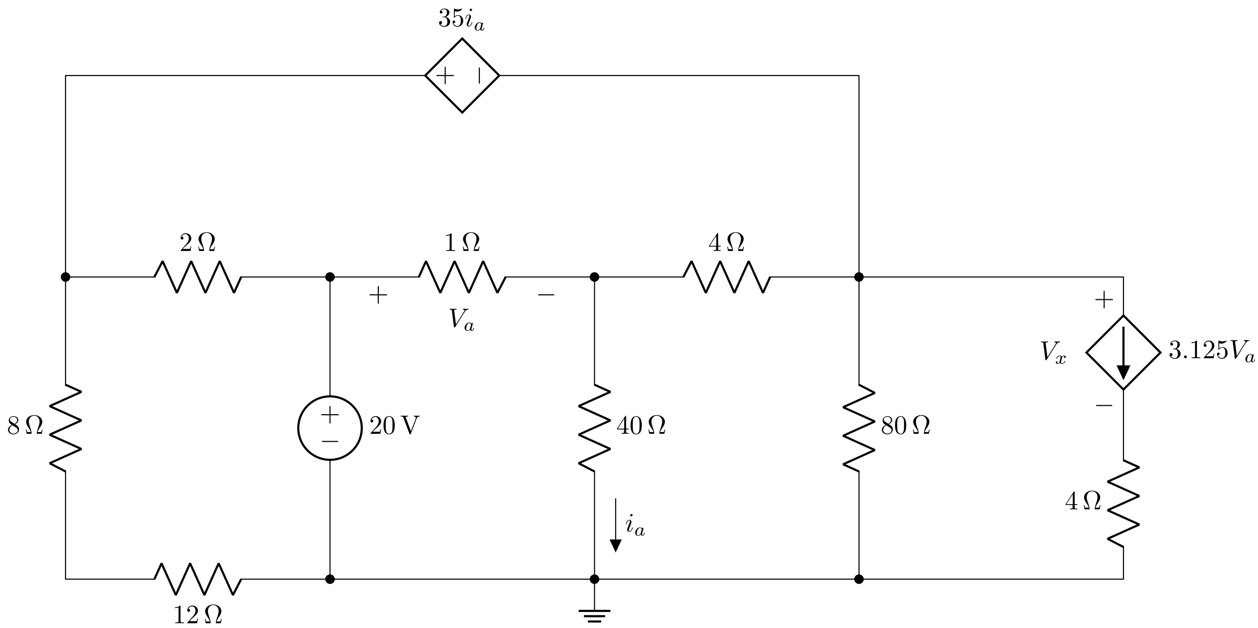
- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to invigilator.
- ✓ All **4 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Use the grid provided on the question paper to plot graphs for questions 4(a) and 4(b).
- ✓ Symbols have their usual meanings.

■ Question 1 of 4 *[CO2] [12 marks]*

Apply Superposition Principle and/or **Source Transformation** to determine the voltage I_x in the following circuit.



■ Question 2 of 4 *[CO3] [15 marks]*

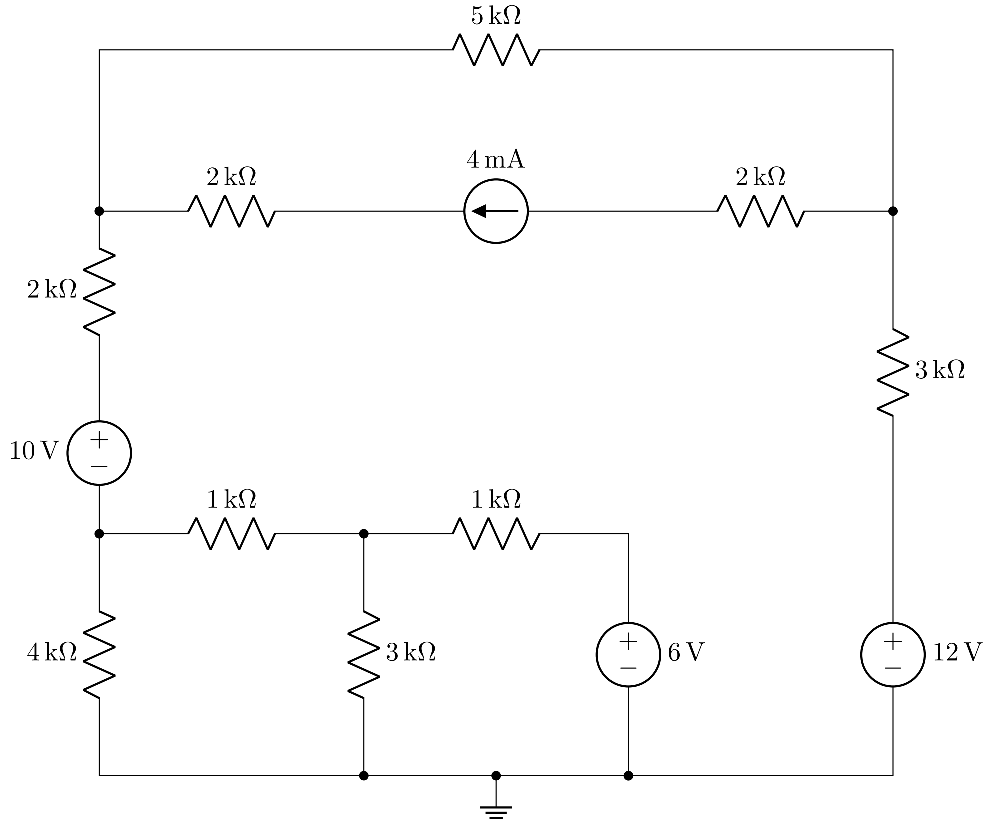


Apply Nodal/Mesh analysis to answer the following questions:

(a) *[12 marks]* Find all the node voltages/mesh currents in the circuit. Note that you need to find either one based on the analysis method you are using.

(b) *[3 marks]* Determine the voltage V_x across the dependent current source as labeled in the circuit diagram.

■ Question 3 of 4 *[CO3] [16 marks]*



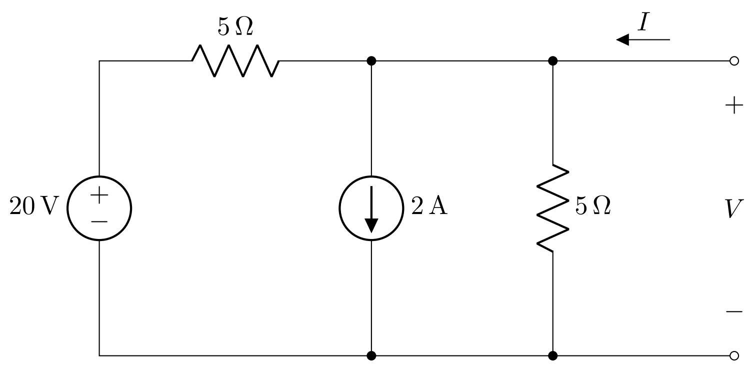
Apply Nodal/Mesh analysis to answer the following questions:

(a) *[12 marks]* Find all the node voltages/mesh currents in the circuit shown above. Note that you need to find either one based on the analysis method you are using.

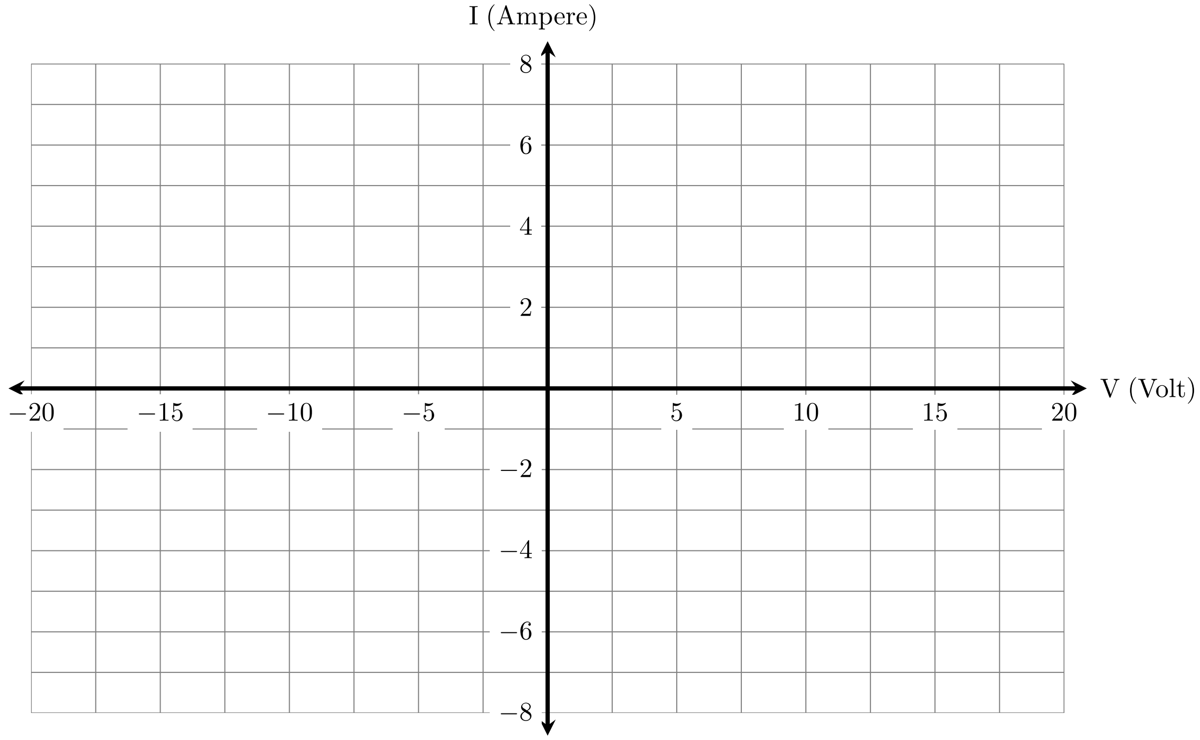
(b) *[4 marks]* Determine the power associated with the 4 A current source and specify whether it is supplying or absorbing power. Make sure to use the appropriate sign and unit.

■ Question 4 of 4 *[CO3] [12 marks]*

(a) *[3 marks]* Plot the $I - V$ characteristics of the following circuit in the grid provided on the next page.



(b) *[3 marks]* Plot the $I - V$ characteristics of the same circuit in the same grid if the 20 V voltage source and the 2 A current source are turned off.



(c) ***[6 marks]*** Determine the equivalent resistance between terminals a and b for the circuit shown below.